



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SUBRAMANYAA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

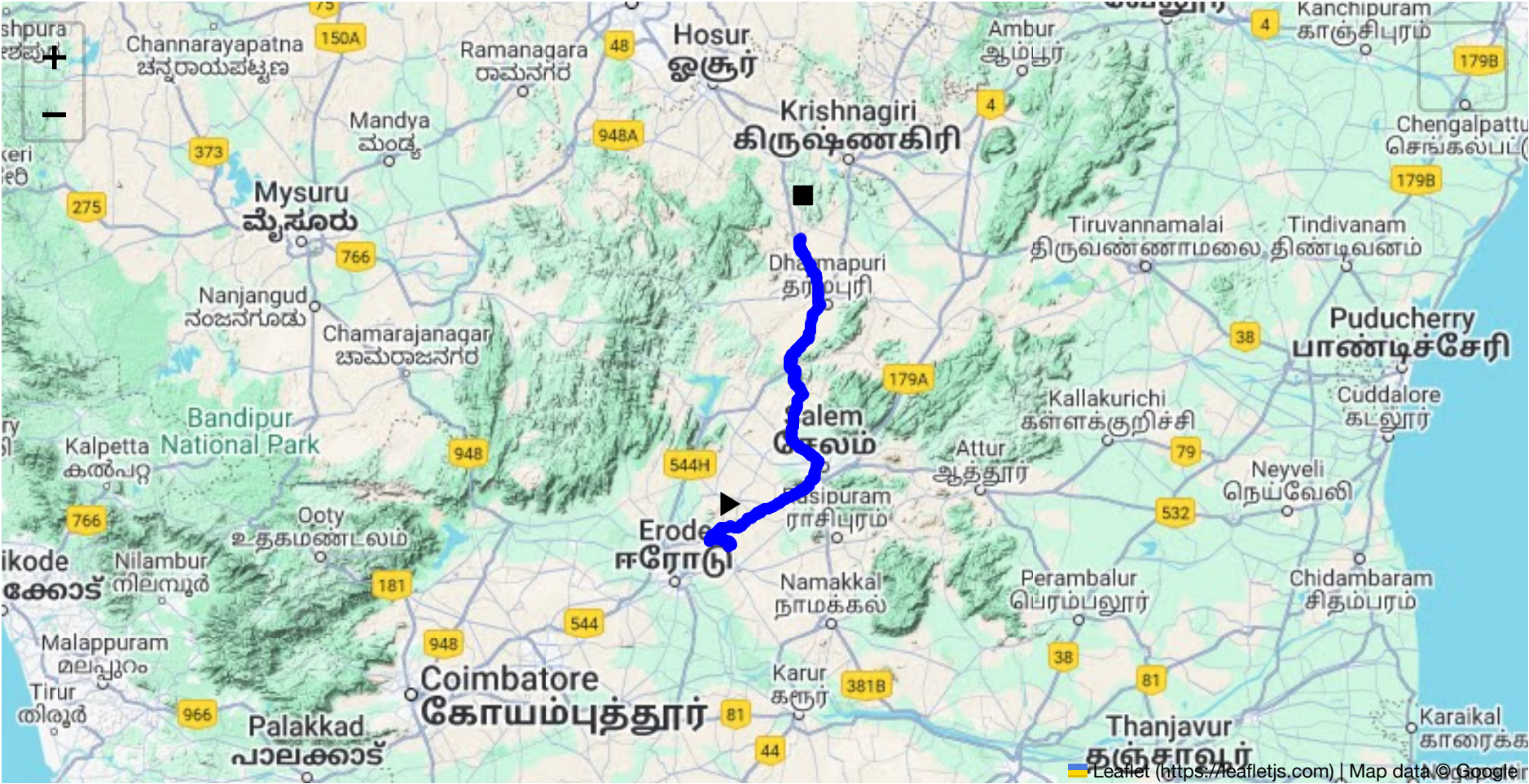
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 136.56 km
Estimated Duration: 2.5 hours
Adjusted Duration (Heavy Vehicle): 3.1 hours

Start: (11.4381, 77.8734)
End: (12.30724, 78.078332)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route starts at Sangagiri, passing through Padaiveedu on Bhavani Main Road, and ends at Palacode. It primarily involves state highways and is approximately 136.56 kilometers long. The journey takes about 2.48 hours for heavy vehicles.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences tropical weather with the possibility of heavy rains during the monsoon season from October to December. This can lead to slippery roads and reduced visibility. It's important to prepare for potential flooding or water-logged roads, especially in low-lying areas.

3. Analysis of Traffic Patterns

Traffic is typically heavier during peak hours, particularly from 8 AM to 10 AM and 5 PM to 8 PM. Congestion can be expected near market areas and junctions, especially in towns and near Sangagiri and Padaiveedu.

4. Assessment of Road Quality and Infrastructure

The road quality varies along the route. While main highways are generally well-maintained, some sections may have potholes, especially in rural stretches. The infrastructure may lack proper signage or lighting in certain areas.

5. Suggestions for Alternative Routes for Emergencies

In case of road blockage or emergencies, consider using alternative state highways that bypass major towns. GPS navigation should be used proactively to find viable detours based on real-time conditions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

In Tamil Nadu, transport of hazardous materials requires adherence to specific regulations, including obtaining proper permits and following designated routes. Vehicle inspection is mandatory to ensure compliance with safety standards.

7. Overview of Historical Incidents

There have been incidents of accidents involving heavy vehicles, particularly due to overloading and brake failures. Data on hazardous materials incidents is limited, but high caution is advised near densely populated areas.

8. Environmental Considerations and Sensitive Areas

The route traverses through some environmentally sensitive areas like agricultural lands and near water bodies. Drivers should be cautious to prevent any spillage or contamination, adhering to containment protocols.

9. Analysis of Communication Coverage

Most parts of the route have decent communication coverage; however, certain rural stretches might have weak signals, causing potential dead zones. It is advisable to equip vehicles with satellite phones or radio communication for emergencies.

10. Estimated Emergency Response Times

Emergency services are generally accessible within 30-60 minutes in towns. Rural stretches may see a response time increase to 1-2 hours due to distance from major facilities.

12. Overall Summary of Risk Assessment

The main risks involve weather-related disruptions, traffic congestion, road quality issues, and regulatory compliance. Proper planning, adherence to regulations, real-time navigation, and emergency protocols are crucial to manage these risks effectively. Training programs for drivers focusing on these elements would be beneficial to ensure their safety and the security of the materials transported.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	High	11.45542, 77.81725	15 KM/Hr
7	Turn	High	11.45636, 77.81678	15 KM/Hr
8	Turn	Medium	11.53573, 77.95794	30 KM/Hr
9	Turn	Medium	11.64183, 78.11810	30 KM/Hr
10	Turn	Medium	11.64338, 78.11998	30 KM/Hr
11	Turn	Medium	11.74068, 78.05047	30 KM/Hr
12	Turn	Medium	11.78039, 78.05792	30 KM/Hr
13	Turn	Medium	11.86753, 78.08331	30 KM/Hr
14	Turn	Medium	11.86981, 78.08387	30 KM/Hr
15	Turn	Medium	11.88041, 78.08143	30 KM/Hr
16	Turn	Medium	11.91302, 78.05856	30 KM/Hr
17	Turn	Medium	11.95625, 78.04748	30 KM/Hr
18	Turn	High	11.95824, 78.04981	15 KM/Hr
19	Turn	High	11.96040, 78.04947	15 KM/Hr
20	Turn	Medium	11.96320, 78.05139	30 KM/Hr
21	Turn	Medium	11.96401, 78.05134	30 KM/Hr
22	Turn	Medium	11.96463, 78.05077	30 KM/Hr
23	Turn	Medium	11.96504, 78.04975	30 KM/Hr
24	Turn	High	11.96597, 78.04920	15 KM/Hr
25	Turn	Medium	11.96679, 78.04942	30 KM/Hr
26	Turn	Medium	11.96804, 78.05080	30 KM/Hr
27	Turn	Medium	12.12067, 78.13281	30 KM/Hr
28	Turn	High	12.26532, 78.08427	15 KM/Hr
29	Turn	High	12.26548, 78.08438	15 KM/Hr
30	Turn	Medium	12.27200, 78.08089	30 KM/Hr
31	Turn	High	12.29579, 78.07370	15 KM/Hr
32	Turn	High	12.29649, 78.07542	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724,	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
			78.0096		
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
11	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
12	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
13	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
14	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
16	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
17	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
18	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
19	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
20	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
23	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
25	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
26	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
29	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
30	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
31	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
33	hospital	Government Hospital Palacode	12.2964824, 78.0761735	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
15	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
21	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
22	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
24	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
27	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
28	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
32	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53573, 77.95794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64338, 78.11998



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74068, 78.05047



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78039, 78.05792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86753, 78.08331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86981, 78.08387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88041, 78.08143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91302, 78.05856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95625, 78.04748



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95824, 78.04981



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96040, 78.04947



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96401, 78.05134



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96463, 78.05077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96504, 78.04975



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96597, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96679, 78.04942



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96804, 78.05080



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12067, 78.13281



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.26532, 78.08427



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.26548, 78.08438



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.27200, 78.08089



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.29579, 78.07370



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.29649, 78.07542
